

VINNYTSIA, Ukraine

WMO: 335620

Lat: 49.2478N

Lon: 28.6042E

Elev: 298

StdP: 97.80

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.2	-21.5	0.6	-18.0	-18.1	0.8	-14.7	12.0	-0.3	10.3	1.3	3.1	0	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.2	19.7	28.4	19.1	26.9	18.4	21.1	27.5	20.3	26.4	19.5	25.2	3.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.3	24.0	18.2	13.5	23.0	17.4	12.9	22.2	62.6	27.7	59.7	26.6	56.9	25.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.2	7.3	DB	-21.9	32.4	3.9	1.9	-24.7	33.7	-27.0	34.8	-29.2	35.9	-32.0	37.3	(4)
(5)			WB	-22.2	22.4	3.8	1.1	-24.9	23.2	-27.2	23.8	-29.3	24.4	-32.1	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-4.3	-2.8	1.9	9.2	14.8	18.1	20.0	19.3	14.1	8.2	2.8	-2.3	(6)	
	DBStd	9.71	5.64	5.67	4.80	4.26	4.04	3.38	3.03	3.35	3.87	4.40	4.80	5.38	(7)	
	HDD10.0	1819	442	358	251	63	8	0	0	0	8	87	218	383	(8)	
	HDD18.3	3846	701	592	508	274	122	47	18	28	135	315	465	641	(9)	
	CDD10.0	1201	0	0	1	40	157	242	309	287	132	31	3	0	(10)	
	CDD18.3	185	0	0	0	1	12	38	69	57	8	0	0	0	(11)	
	CDH23.3	1598	0	0	0	9	116	297	567	529	79	2	0	0	(12)	
	CDH26.7	390	0	0	0	1	16	51	152	157	13	0	0	0	(13)	
(14) Wind	WSAvg	3.9	4.1	4.3	4.4	4.1	3.8	3.6	3.4	3.2	3.5	3.7	4.1	4.2	(14)	
(15) Precipitation	PrecAvg	632	38	35	33	46	58	82	98	59	52	38	45	41	(15)	
	PrecMax	857	150	92	83	120	134	193	246	125	182	100	151	80	(16)	
	PrecMin	429	7	7	2	10	13	14	13	16	1	3	5	7	(17)	
	PrecStd	100	25	22	21	28	30	49	52	32	44	27	28	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	10.5	16.3	24.2	28.7	29.9	32.6	32.7	28.2	22.6	15.9	9.9	(19)	
		MCWB	5.1	6.9	9.6	14.4	17.5	19.6	20.5	20.5	17.3	15.1	11.5	7.9	(20)	
	2%	DB	5.0	7.1	13.3	21.1	26.3	28.1	30.3	30.3	25.7	19.7	12.9	7.4	(21)	
		MCWB	3.7	5.0	7.9	12.6	16.5	19.1	20.0	19.7	16.3	14.0	10.3	6.2	(22)	
	5%	DB	3.4	5.3	10.8	18.6	24.3	26.7	28.2	28.2	23.4	16.8	11.0	5.7	(23)	
		MCWB	2.3	3.7	6.5	11.3	16.0	18.6	19.7	18.9	15.6	12.7	9.5	4.8	(24)	
	10%	DB	2.1	3.5	8.6	16.6	22.3	25.0	26.5	26.3	21.1	14.5	9.3	3.9	(25)	
		MCWB	1.3	2.3	5.4	10.2	14.9	17.9	18.9	18.3	14.9	11.5	8.0	3.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	7.2	10.3	14.8	19.1	21.9	22.1	22.2	19.1	16.4	12.4	8.5	(27)	
		MCDB	6.5	9.5	15.2	23.0	25.7	26.9	28.7	29.2	25.6	20.9	14.7	9.6	(28)	
	2%	WB	3.8	5.1	8.5	13.5	17.9	20.5	21.2	20.9	17.4	14.6	10.8	6.3	(29)	
		MCDB	4.8	6.7	12.2	19.8	24.2	26.0	27.7	27.7	23.1	18.1	12.3	7.3	(30)	
	5%	WB	2.4	3.8	7.0	12.1	16.7	19.6	20.5	20.0	16.3	13.1	9.5	4.8	(31)	
		MCDB	3.2	5.2	9.9	17.1	22.6	24.8	26.6	26.4	21.5	16.2	10.8	5.6	(32)	
	10%	WB	1.3	2.4	5.6	10.9	15.6	18.5	19.6	19.0	15.3	11.7	8.1	3.1	(33)	
		MCDB	2.0	3.3	8.2	15.3	21.0	23.5	25.3	24.8	20.1	14.1	9.3	3.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.0	6.9	9.2	10.1	9.6	9.8	10.4	9.0	7.6	4.9	4.3	(35)	
		MCDBR	4.5	6.5	10.8	12.3	12.5	11.8	12.5	13.1	12.9	11.2	7.5	5.4	(36)	
	5% WB	MCWBR	3.9	5.0	6.4	6.0	5.3	5.0	4.7	5.1	5.6	6.3	5.1	4.5	(37)	
		MCDWR	4.2	6.2	9.7	11.4	11.3	10.6	11.0	11.6	10.8	9.9	6.5	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.338	0.375	0.434	0.404	0.401	0.429	0.430	0.392	0.360	0.339	0.307	(40)	
		taud	2.234	2.186	2.203	2.112	2.272	2.298	2.230	2.221	2.291	2.357	2.327	2.292	(41)	
	Edn at Noon		724	788	824	810	857	860	827	805	799	762	685	674	(42)	
		Edh at Noon	75	101	118	144	128	126	133	128	108	85	70	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.96	1.74	3.02	4.41	5.73	6.06	5.83	5.16	3.54	2.15	0.92	0.67	(44)	
	RadStd		0.18	0.34	0.43	0.46	0.46	0.42	0.37	0.30	0.53	0.32	0.10	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.78	N/A	N/A	+0.86	+0.66	+0.56	-165	-242	+136	+50
Regional (11 neighbors)		+0.72	N/A	N/A	+0.90	+0.49	N/A	-136	-202	+119	+51

Nomenclature: See separate page